Task 3. Research Results, Data Analysis and Visualization

Comparative Evaluation of Machine Learning Models for Phishing URL Detection

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GitHub: https://github.com/dariyanagabdylshukir-crypto/phishing-url-ml-comparison

1. Goal and experiment

We compare Logistic Regression (baseline), Random Forest, SVM and XGBoost for phishing URL detection. Each URL is described by 16 lexical features (length, dots, digits, keywords); the selected set uses 7 of them.

Dataset: public labelled URL dataset (faizann24 on GitHub), duplicates removed, stratified sample of 20 000 URLs, 16% phishing. We ran 4 models × 2 feature sets × 10 repeats (70/30 split, seeds 42–51). Latency was measured for 1, 10, 100 and 1000 URLs per request; in each series the first 5 calls are warm-up and were removed, 20 calls were used. The raw log is data/raw/test_run.jsonl, the analysis is in notebooks/01_data_analysis.ipynb.

Statistics: Mean ± SD, Median (p25–p75), p95, p99 and 95% bootstrap CI (10 000 resamples, seed 42). Guardrails: error rate = 0, recall ≥ 0.80, p99 latency ≤ 100 ms.

2. Domain model

The domain model (loss function, metrics and notation) is in docs/domain_model.md. The main metric is the F1-score for the phishing class.

3. Results

Model	Features	Role	F1 Mean ± SD	F1 95% CI	Recall Mean ± SD	Latency 1 URL, ms Median (p25–p75)	p99, ms
Logistic Regression	full	**Baseline**	0.314 ± 0.015	[0.305, 0.323]	0.193 ± 0.012	0.33 (0.28–0.40)	0.69
Random Forest	full	Proposed	0.638 ± 0.011	[0.632, 0.645]	0.557 ± 0.010	4.66 (4.19–5.13)	6.45
SVM	full	Proposed	0.403 ± 0.020	[0.391, 0.414]	0.260 ± 0.016	0.55 (0.49–0.59)	0.95
XGBoost	full	Proposed	0.626 ± 0.016	[0.617, 0.635]	0.504 ± 0.019	0.40 (0.34–0.47)	0.74
Logistic Regression	selected	Proposed	0.226 ± 0.018	[0.215, 0.236]	0.129 ± 0.011	0.31 (0.27–0.36)	0.63
Random Forest	selected	Proposed	0.400 ± 0.019	[0.389, 0.412]	0.300 ± 0.019	4.49 (3.91–5.10)	8.38
SVM	selected	Proposed	0.287 ± 0.017	[0.277, 0.297]	0.171 ± 0.012	0.47 (0.41–0.51)	0.76
XGBoost	selected	Proposed	0.385 ± 0.014	[0.377, 0.393]	0.256 ± 0.013	0.40 (0.35–0.44)	0.93

Table 1. Results (10 repeats). All configurations have error rate 0 and p99 below 10 ms; none reaches recall 0.80.

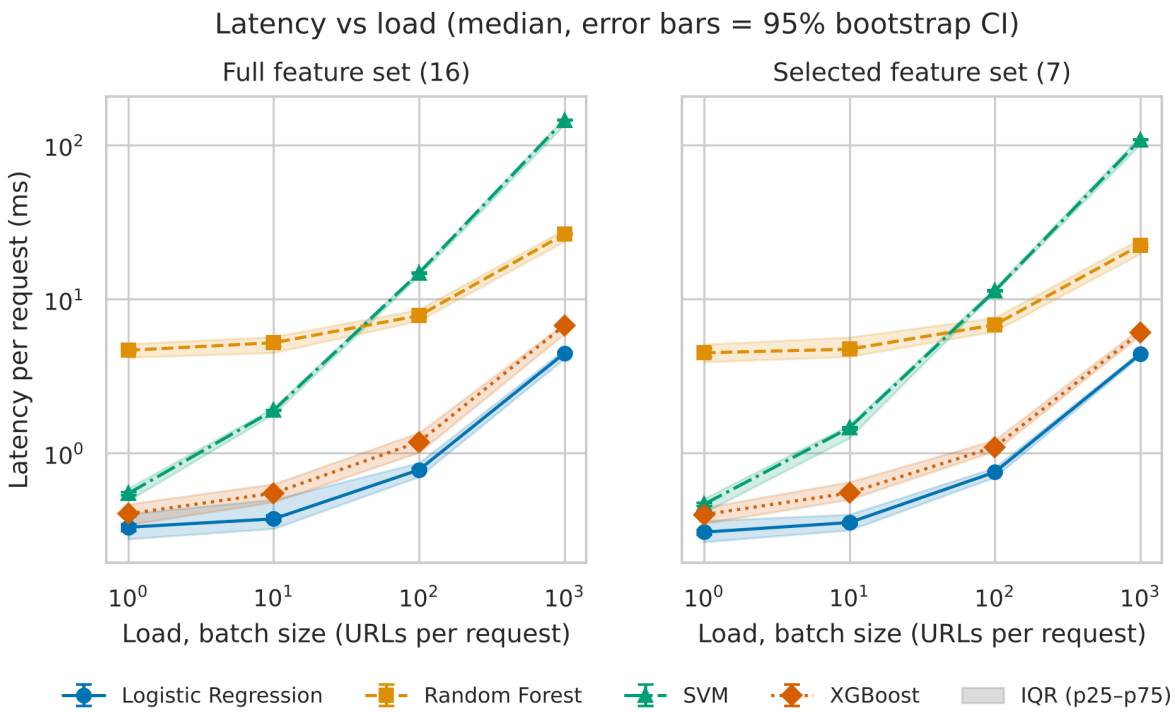


Figure 1. Latency vs load (log scale). Error bars: 95% CI of the median, band: p25–p75.

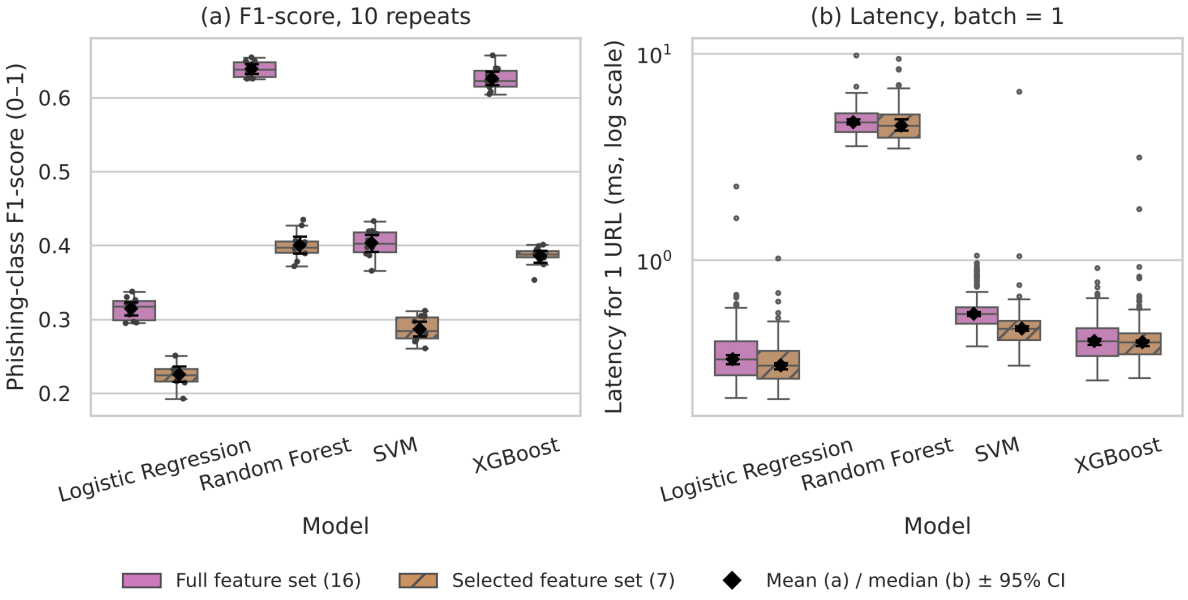


Figure 2. (a) F1 over 10 repeats, (b) latency for one URL. Diamonds: mean / median with 95% CI.

4. Discussion

RQ1. Random Forest (F1 = 0.64) and XGBoost (0.63) are about two times better than the baseline Logistic Regression (0.31), and their confidence intervals do not overlap with the baseline. Accuracy (0.85–0.90) is not useful here, because predicting "benign" for every URL already gives 0.84.

RQ2a. With 7 features F1 decreases for every model (Random Forest 0.64 → 0.40), so H0 is rejected.

RQ2b. Latency changes by less than 0.2 ms, because most of the time is spent on feature extraction. The small feature set does not give a real speed-up.

RQ3. Logistic Regression and XGBoost are the fastest (0.3–0.4 ms per URL), Random Forest needs 4.7 ms, SVM is very slow for large batches (more than 100 ms for 1000 URLs). XGBoost with all features has the best balance.

Guardrails. The best recall is 0.56 (Random Forest), so no model is ready for real use. Simple lexical features are not enough; next steps are class weights, a lower threshold or more features.

Limitations. The dataset labels malicious URLs in general, default model parameters were used, latency depends on the computer.

5. Where to find

What	File
Raw log (warm-up and evaluation calls)	data/raw/test_run.jsonl
Aggregated metrics	data/processed/metrics_summary.csv
Processing and statistics	src/process_data.py, notebook sections 1–2
Figures (pdf + png 300 dpi)	reports/figures/, src/visualize.py, notebook section 3
Domain model	docs/domain_model.md
Experiment and dataset scripts	src/run_experiment.py, src/prepare_dataset.py, configs/experiment.yaml

6. Author contributions

Author	Task 2	Task 3
Dariyana Gabdylshukir	methodology, Docker pipeline, smoke test	notebook, domain model, README results, report
Zhanna Azatkyzy	repository upload	dataset and experiment runs, data processing, figures, Git and release